Latent Pathways and Post-Disruption Reorganization in Mexican Cartel Networks

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Abstract

This paper analyzes the BACRIM 2020 signed network of 150 Mexican criminal organizations using centrality analysis, null-model comparison, link prediction, and targeted node-removal simulation. The alliance network is sparse, clustered above random baselines, and dominated by two hubs (CJNG and Sinaloa). A dyadic logistic model incorporating geography, shared rivalries, and specialty outperforms topology-only link prediction (AUC 0.73 vs. chance). Targeted removal of CJNG fragments the network into multiple disconnected components, consistent with theoretical predictions for heavy-tailed topologies. Events between 2024 and 2026, including the arrest of El Mayo Zambada, the killing of El Mencho, and mass extraditions, track the structural vulnerabilities flagged by this static 2020 snapshot.

1 Introduction

1.1 Context and Motivation

Organized crime is a relational phenomenon. Alliances and rivalries among criminal organizations shape who controls territory, who moves product, and where violence concentrates. In Mexico, drug cartels form webs of cooperation and competition that have contributed to over 290,000 homicides and 175,000 disappearances between 2012 and 2021 (INEGI, 2022). Understanding the structure of these webs is presented here as an academic exercise grounded in real policing challenges, with potential relevance for disruption strategy, intelligence prioritization, and policy design.

The BACRIM 2020 dataset (Prieto-Curiel and Campedelli, 2023) provides a strong empirical base, a signed network of 150 Mexican criminal organizations where positive ties denote alliances and negative ties denote rivalries. Yet a static map alone cannot answer key analytical questions. Is the observed structure systematic or random? Which currently disconnected pairs are most likely to form future alliances? How does the network change when high-centrality actors are removed?

Prior work provides the methodological foundation for addressing these questions. Network studies of criminal organizations emphasize brokerage, clustering, and degree heterogeneity as core structural features (Morselli, 2009; Sparrow, 1991). Signed-network research formalizes balance mechanisms relevant to alliance-rivalry systems (Heider, 1946; Cartwright and Harary, 1956; Leskovec et al., 2010). Robustness literature explains why hub-targeted interventions can produce disproportionate fragmentation (Albert et al., 2000; Duijn et al., 2014). Link-prediction research defines the topological metrics used to infer likely future ties (Liben-Nowell and Kleinberg, 2007; Berlusconi et al., 2016). For Mexico specifically, BACRIM and related studies provide empirical context on territorial competition and fragmentation dynamics (Prieto-Curiel and Campedelli, 2023; Coscia and Rios, 2012; Calderón et al., 2015).

Our contribution is to analyze a contemporary cartel network dataset, enriched with specialty attributes from DEA assessments and open-source intelligence, using graph-topological analysis, null-model benchmarking, alliance link prediction, and targeted-disruption simulations to address the following research problems:

RP1: What structural properties characterize the cartel alliance network, and how does it differ from random baselines?

RP2: Which cartel pairs are most likely to form future alliances, and do non-topological features (geography, rivalry, specialty) improve prediction over topology alone?

RP3: How does targeted removal of central hubs fragment the network, and how does the surviving network plausibly reorganize?

This paper is organized as follows. Section 2 describes the data and contextual overview. Section 3 presents the analytical methods. Section 4 reports findings on network structure,

alliance prediction, and disruption scenarios. Section 5 interprets these results in light of recent developments, and Section 6 concludes with limitations and future directions.

2 Data and Context

2.1 Collection and Preprocessing

We utilize the BACRIM 2020 dataset (Prieto-Curiel and Campedelli, 2023), a publicly available criminal-network dataset that provides node and edge information for Mexican cartel organizations. The dataset includes three primary files:

- `BACRIM2020_Nodes.csv` — (node data) 150 cartel organizations with group name, state, and short name.
- `BACRIM2020_Alliances.csv` — (edge data) alliance ties with positive weights representing the number of Mexican states where the alliance is active.
- `BACRIM2020_Rivals.csv` — (edge data) rivalry ties with negative weights.

Because BACRIM 2020 provides the core network structure but not all attributes required for our analyses (e.g., organizational specialties and broader violence context), we supplement it with the following additional sources:

- `BACRIM2020_Nodes_With_Specialties.csv` — (node data) node-level content that extends BACRIM organizations with organizational type (National Cartel, Regional Cartel, Local, Cell, Toll-Collector), primary specialty, activity list, and drug portfolio. These specialty attributes were compiled from multiple external references and harmonized to BACRIM entities, with row-level source attribution (e.g., DEA NDTA 2024/2025, CRS report R41576, and InSight Crime profiles (U.S. Drug Enforcement Administration, 2024, 2025; Congressional Research Service, 2024; InSight Crime, 2026)).
- `Trends2012_2021.csv` — (context data, non-network) annual counts of homicides, missing persons, and arrests in Mexico from INEGI (INEGI, 2022).

To build the dataset, we first combine the two BACRIM 2020 edge datasets (alliances and rivalries). Each record is mapped to a common organization list, producing a unified tie table with a consistent node reference scheme. Next, we remove duplicate edges by writing each undirected pair in a fixed order $(\min(u, v), \max(u, v))$, so repeated or reversed pairs are treated as the same relationship. Isolate nodes (organizations with no alliance ties) are removed from the alliance-only network for structural analysis. Finally, we construct two graphs:

1. $G_{\textbf{signed}}$ (Signed Master Graph): an undirected graph where each alliance edge carries weight $+1$ and each rivalry edge carries weight -1.

2. $G_{\textbf{alliance}}$ (Alliance Subgraph): the positive-edge subgraph used for centrality, link prediction, and resilience tasks.

All code, data, and outputs required to replicate this study are available at https://github.com/ElvisCasco/cartel-alliance-prediction.

2.2 Contextual Overview

The BACRIM 2020 network is a snapshot from a period of historically high violence. INEGI data records a sharp escalation beginning in 2017, coinciding with the intensification of Mexico's kingpin strategy and the fragmentation of major cartels. The 2020 snapshot captures the network at roughly 36,800 annual homicides and 22,900 disappearances, near-peak levels. Network fragmentation driven by law enforcement operations and inter-cartel competition fuels this violence directly, as splinter groups fight for territorial control (Calderón et al., 2015). Figure 1 plots these trends with the BACRIM snapshot year marked.

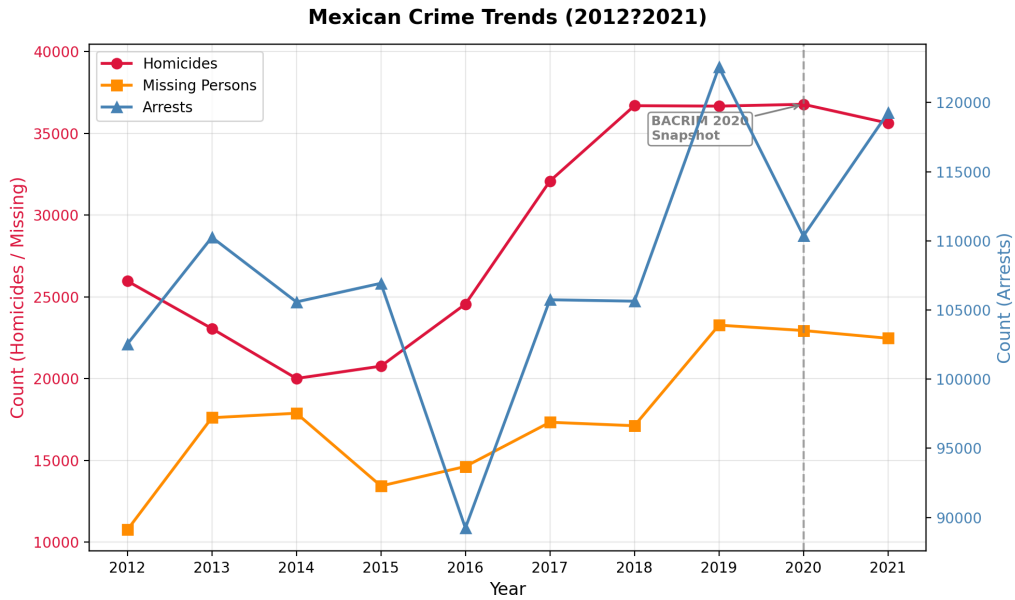

Figure 1: Mexican crime trends, 2012–2021. The vertical dashed line marks the BACRIM 2020 snapshot year. Source: INEGI (2022).

The 150 organizations vary widely in scale and function, from national cartels operating across multiple states to local cells controlling a single neighborhood. Drug trafficking dominates as the primary specialty, but organizations also specialize in extortion, fuel theft, and drug retail. Methamphetamine and fentanyl are the most common drug portfolio entries, consistent with DEA 2024–2025 assessments. Figure 2 summarizes this composition.

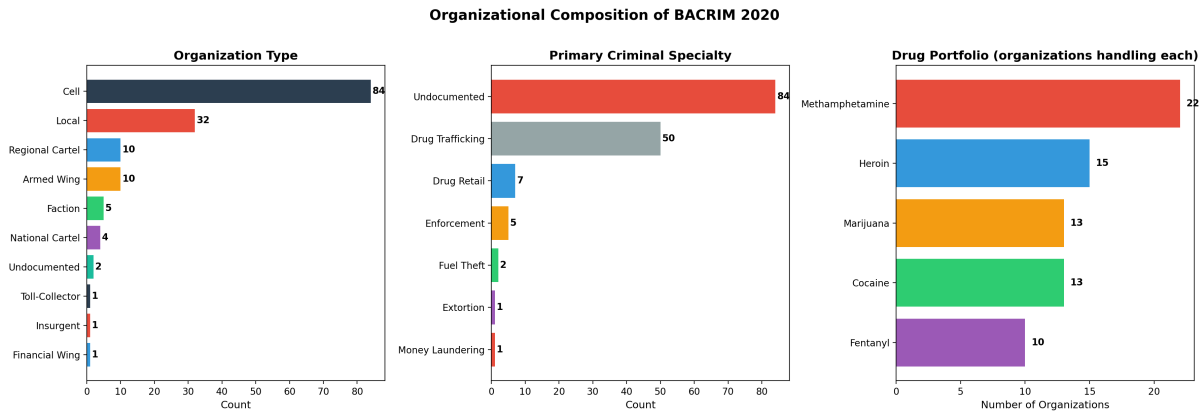

Figure 2: Organizational composition of the BACRIM 2020 network: type distribution, primary criminal specialty, and drug portfolio.

Of these 150 organizations, 91 have at least one alliance tie, forming G_{alliance} with exactly 91 edges, barely 2% of all possible alliances (Table 1). The network splits into 8 components; the largest holds 50 nodes (55%), with diameter 8 and average path length 3.30. The global clustering coefficient ($C = 0.036$) is low in absolute terms but, as we show in Section **??**, considerably higher than random baselines.

Table 1: Basic statistics of the alliance network.

Metric	**Value**
Nodes (N)	91
Edges (E)	91
Density (ρ)	0.0222
Global Clustering Coefficient (C)	0.0356
Diameter (LCC)	8
Average Path Length (LCC)	3.30
Connected Components	8
Largest Component Size	50

Organizations concentrate in a handful of states (Jalisco, Michoacán, Guerrero, Tamaulipas, and Sinaloa) and the alliance edges split roughly evenly between same-state and cross-state ties (Figure 3). Same-state alliances reflect territorial cooperation, while cross-state alliances suggest supply-chain logic, with production hubs in one state allied with distribution nodes in another. Both proximity and long-range logistics shape the cooperation structure.

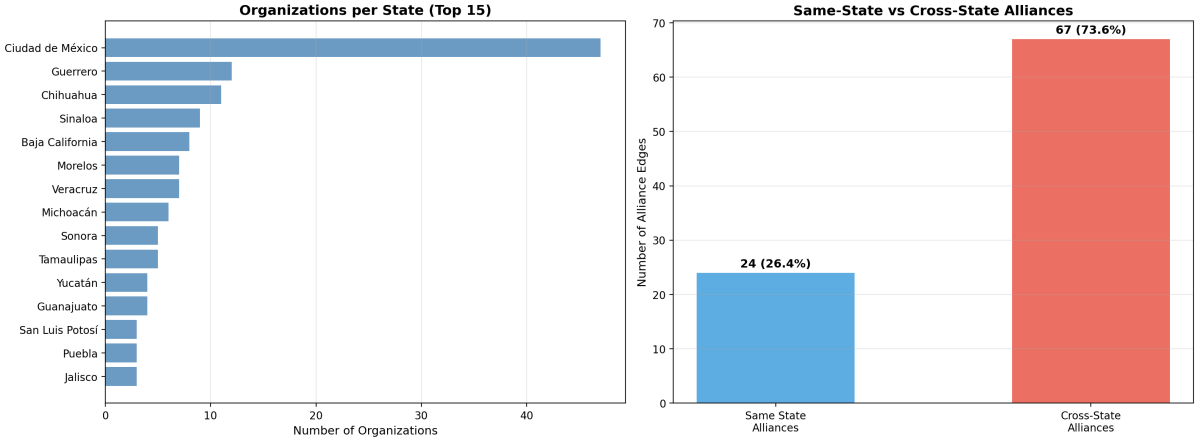

Figure 3: Left: number of organizations per state (top 15). Right: same-state vs. cross-state alliances.

Figure 4 separates alliance and rivalry edges into side-by-side panels using a shared force-directed layout. In the alliance panel, national cartels sit near the center with many local cells connected through a single edge to a hub, organizations that would become isolates upon its removal. CJNG's allies are scattered across multiple geographic clusters, while Sinaloa's are more geographically concentrated. In the rivalry panel, edges run between clusters rather than within them, consistent with structural balance theory (Cartwright and Harary, 1956). The separation reveals a bipolar structure in which the two dominant hubs compete for the same satellite organizations, and most smaller groups are allied with one hub and rival to the other.

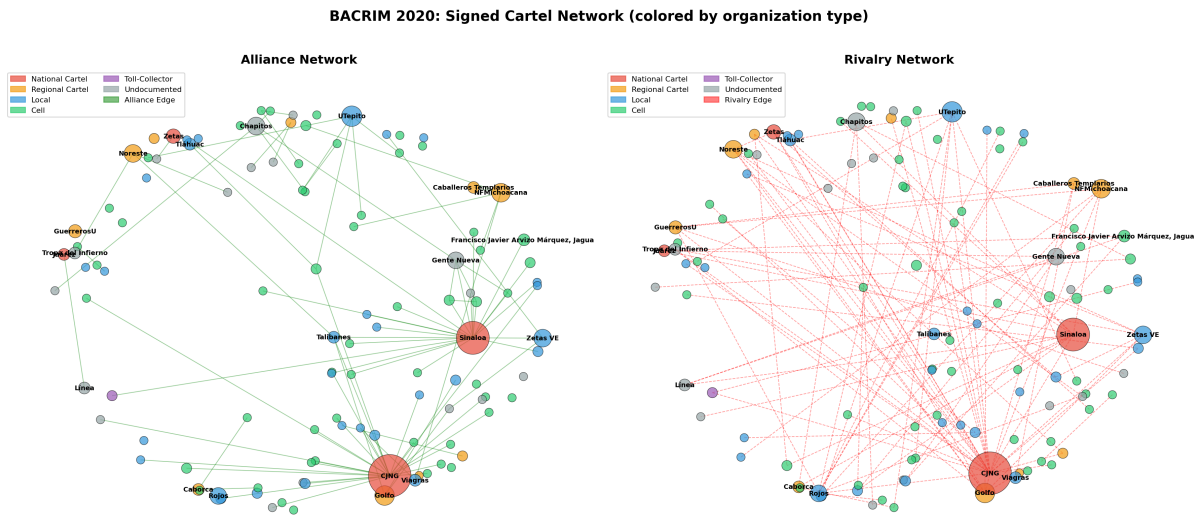

Figure 4: BACRIM 2020 network separated by edge type. Left: alliance edges (green). Right: rivalry edges (red, dashed). Node color indicates organization type; node size is proportional to degree. Both panels share the same node positions.

Three patterns in this overview shape the methodology that follows. First, the network is sparse and hub-dominated, with connectivity depending on a few central nodes, which motivates centrality analysis and targeted-removal simulations. Second, the non-trivial clustering suggests

that alliances are not random, calling for null-model benchmarking. Third, the geographic concentration and rivalry structure indicate that topology alone may not explain alliance formation, motivating a dyadic prediction model that incorporates geography, shared enemies, and organizational specialty alongside network features.

3 Methods

All three research problems concern the cooperation structure, so we take G_{alliance} as the primary object of analysis. Rivalries enter as predictive features rather than edges to be modeled, since post-disruption reorganization is driven by new alliances, not new enemies. Figure 5 summarizes the pipeline.

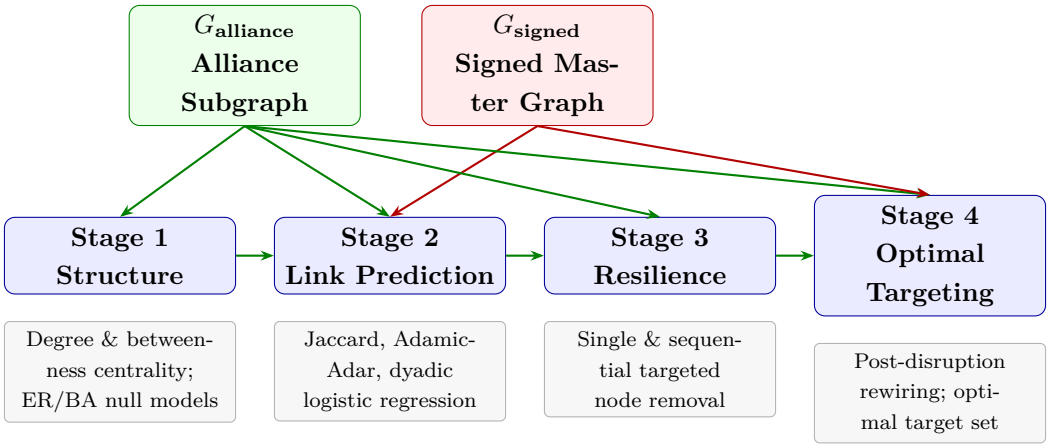

Figure 5: Analytical pipeline. G_{alliance} feeds all four stages; G_{signed} supplies rivalry-based features to the logistic model in Stages 2 and 4.

3.1 Stage 1: Structural Characterization

Addressing RP1 requires identifying which organizations occupy structurally critical positions and determining whether the observed topology departs from random baselines. We apply centrality analysis for the former and null-model comparison for the latter.

3.1.1 Centrality Measures

We compute two centrality measures on G_{alliance}. Degree centrality identifies hubs (many alliances), while betweenness centrality identifies brokers (nodes that bridge otherwise disconnected clusters, even if their own alliance count is modest).

For a node v in an undirected graph with N nodes, degree centrality normalizes the number of incident edges by the maximum possible:

$$C_D(v) = \frac{k_v}{N-1} \tag{1}$$

where $k_v = \sum_j A_{vj}$. Betweenness centrality measures how often a node lies on shortest paths between other pairs:

$$C_B(v) = \sum_{\substack{s \neq v \neq t \\ s,t \in V}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the total number of shortest paths from s to t and $\sigma_{st}(v)$ is the number passing through v. We normalize by $\frac{(N-1)(N-2)}{2}$ and use Brandes' algorithm (Brandes, 2001).

3.1.2 Null Models

Sparsity or clustering is only informative if it differs from what random wiring would produce. We benchmark G_{alliance} against two null models matched on N and expected E, one based on purely random attachment (Erdős–Rényi) and one based on preferential attachment to high-degree nodes (Barabási–Albert).

In the Erdős–Rényi $G(N, p)$ model, each of the $\binom{N}{2}$ possible edges exists independently with probability $p = 2E_{\text{real}}/(N(N-1))$, producing a Poisson degree distribution and expected clustering $C_{\text{ER}} = p$ (Erdős and Rényi, 1959). In the Barabási–Albert model, each new node attaches to m existing nodes with probability proportional to current degree, producing a power-law distribution $P(k) \sim k^{-3}$; we set $m = \text{round}(E/N)$ to match the average degree (Barabási and Albert, 1999). Following Watts and Strogatz (1998), we compute the small-world coefficient:

$$\sigma = \frac{C_{\text{real}}/C_{\text{ER}}}{L_{\text{real}}/L_{\text{ER}}} \tag{3}$$

where C is the global clustering coefficient and L is the average shortest path length in the largest connected component. A value $\sigma > 1$ indicates small-world properties: high local clustering combined with short global paths.

3.2 Stage 2: Link Prediction

Given that the alliance structure is not random, the forces that shaped existing ties should help predict future ones. The outcome we predict is alliance formation, whether a currently unconnected pair will form a cooperative tie. Rivalry information enters not as an outcome but as a predictor, with shared enemies and signed-balance scores from G_{signed} serving as features in the logistic model. We first apply two topological similarity metrics on G_{alliance} alone, then fit a dyadic logistic model that adds rivalry-derived features alongside geography and specialty. If the topological metrics already perform well, non-network attributes add little value. If they do not, contextual factors are necessary and a richer model is justified.

3.2.1 Topological Metrics

We predict future alliances among currently non-adjacent pairs using two topological similarity metrics (Liben-Nowell and Kleinberg, 2007). The Jaccard coefficient measures the fraction of

the combined neighborhood that is shared:

$$J(u, v) = \frac{|\mathcal{N}(u) \cap \mathcal{N}(v)|}{|\mathcal{N}(u) \cup \mathcal{N}(v)|} \tag{4}$$

The Adamic-Adar index weights each shared neighbor inversely by the log of its degree, so that low-degree shared neighbors contribute more (Adamic and Adar, 2003):

$$AA(u, v) = \sum_{w \in \mathcal{N}(u) \cap \mathcal{N}(v)} \frac{1}{\log |\mathcal{N}(w)|} \tag{5}$$

3.2.2 Dyadic Logistic Model

Topology-only metrics ignore geography, shared enemies, and operational complementarity. To test whether these factors improve prediction, we treat each potential pair (i, j) as a binary outcome and fit a logistic regression:

$$P(\text{alliance} \mid i, j) = \frac{1}{1 + e^{-\eta_{ij}}} \tag{6}$$

We define six dyadic features, summarized in Table 2 and detailed below.

Table 2: Dyadic features for alliance prediction.

Symbol	Feature	Source	Type	Mechanism
x_1	SharedAllies	G_{alliance}	Count	Triadic closure
x_2	SharedRivals	G_{signed} (rivalry edges)	Count	Structural balance
x_3	Complementarity	Node attributes	Binary	Specialty heterophily
x_4	SameState	Node attributes	Binary	Geographic proximity
x_5	BalanceScore	G_{signed} (all edges)	Integer	Heider balance
x_6	DegreeProduct	G_{alliance}	Count	Preferential attachment

x_1 (SharedAllies). $|\mathcal{N}_{\text{alliance}}(i) \cap \mathcal{N}_{\text{alliance}}(j)|$, the number of organizations allied with both i and j. Triadic closure predicts that pairs sharing many allies face social pressure toward direct cooperation.

x_2 (SharedRivals). $|\mathcal{N}_{\text{rivalry}}(i) \cap \mathcal{N}_{\text{rivalry}}(j)|$, the number of organizations that are rivals of both i and j, computed on the rivalry-only subgraph extracted from G_{signed}. Structural balance theory predicts that shared enemies push organizations toward alliance ("the enemy of my enemy is my friend").

x_3 (Complementarity). A binary indicator equal to 1 if i and j have different documented primary specialties, 0 otherwise. Pairs where either organization has an undocumented specialty are coded 0. The primary specialties in the dataset are Drug Trafficking, Drug Retail, Extortion, and Fuel Theft. This feature tests whether organizations with different criminal profiles are more likely to ally, for example a trafficking organization pairing with an extortion group to secure territory.

x_4 (SameState). A binary indicator equal to 1 if both organizations operate in the same Mexican state. Geographic proximity lowers coordination costs and creates shared territorial incentives.

x_5 (BalanceScore). For each common neighbor w in G_{signed}, we add $+1$ if both edges (i, w) and (j, w) carry the same sign (both alliances or both rivalries), and -1 if they carry opposite signs. The sum captures Heider balance. Triads where all three pairs are consistent ($+/+/+$ or $-/-/+$) contribute positively, while imbalanced triads contribute negatively.

x_6 (DegreeProduct). $\deg_{\text{alliance}}(i) \times \deg_{\text{alliance}}(j)$, the product of alliance-graph degrees. This operationalizes preferential attachment, where highly connected organizations are more likely to form new ties with each other.

We fit two nested specifications to isolate the contribution of non-topological features:

$$\text{Topology-only:} \quad \eta_{ij}^{\text{topo}} = \beta_0 + \beta_1 x_1 + \beta_6 x_6 \tag{7}$$

$$\text{Full model:} \quad \eta_{ij}^{\text{full}} = \beta_0 + \sum_{k=1}^{6} \beta_k x_k \tag{8}$$

The topology-only model uses SharedAllies and DegreeProduct, features derived entirely from G_{alliance}. The full model adds rivalry, geographic, and specialty features from G_{signed} and node attributes. With 91 edges among $\binom{91}{2} = 4{,}095$ possible pairs (2.2% positive rate), we use class-weighted logistic regression to prevent trivial all-zero predictions.

3.2.3 Evaluation

To compare the four models (Jaccard, Adamic-Adar, topology-only logistic, full logistic), we use 5-fold edge-holdout cross-validation. The 91 alliance edges are randomly partitioned into five folds. In each iteration, one fold (20% of edges) is held out and removed from G_{alliance}, and all four models are trained on the remaining graph. Features are recomputed on this reduced graph so that no model has access to the held-out edges during training. For evaluation, the held-out edges serve as true positives and are paired with an equal number of randomly sampled non-edges as true negatives. Each model scores the test pairs, with Jaccard and Adamic-Adar producing raw similarity values and the logistic models producing predicted probabilities. We measure performance using AUC-ROC, where 0.5 corresponds to random guessing and 1.0 to perfect separation of real alliances from non-edges, and report the mean and standard deviation across folds.

3.3 Stage 3: Network Resilience

The centrality analysis shows that connectivity depends on a handful of hubs. We now ask what happens when those hubs are removed. We simulate targeted node removal on G_{alliance}, first a single hub, then the top hubs sequentially, to measure how quickly the cooperation structure fragments (RP3). Rivalry edges are excluded because we are interested in how the alliance network breaks apart, not in the rivalry topology.

Algorithm 1 describes single targeted removal and Algorithm 2 describes sequential adaptive attack, where degree is recalculated after each removal to reflect the evolving topology. This adaptive approach is more realistic than simultaneous removal since law enforcement operations unfold sequentially.

Algorithm 1 Single targeted removal

Require: G_{alliance}

$v^* \leftarrow \arg\max_v C_D(v)$ ▷ highest-degree node

Record pre-removal metrics (N, E, ρ, C, components, LCC, diameter, L)

Remove v^* and all incident edges from G_{alliance}

Prune new isolates

Record post-removal metrics

return Δ between pre- and post-removal metrics

Algorithm 2 Sequential adaptive attack

Require: G_{alliance}, budget k

$G_0 \leftarrow G_{\text{alliance}}$

for $t = 1, \ldots, k$ **do**

  $v_t \leftarrow \arg\max_{v \in G_{t-1}} \deg_{G_{t-1}}(v)$ ▷ recalculate degree

  Remove v_t and all incident edges from G_{t-1}

  Prune new isolates to obtain G_t

  Record LCC size and component count of G_t

end for

Albert et al. (2000) showed that scale-free-like networks are robust to random failure but fragile under targeted attack. The critical removal fraction is $f_c = 1 - 1/(\kappa - 1)$, where $\kappa = \langle k^2 \rangle / \langle k \rangle$; for heavy-tailed distributions, κ is large and targeted removal rapidly causes collapse. We equate node removal with organizational neutralization, use the alliance-only graph, assume no rewiring after removal, prune isolates, and treat edges as unweighted.

3.4 Stage 4: Optimal Targeting Under Rewiring

Stage 3 holds the network static after removal, but surviving organizations seek new alliances to replace lost connections. A target that fragments the static network may be suboptimal if survivors rewire around the gap. We therefore combine the removal framework with the logistic model from Stage 2: for each candidate target set, we remove the nodes from G_{alliance}, predict new alliances using the model (which draws on rivalry features from G_{signed}), and evaluate the post-rewiring fragmentation.

We formalize this as a combinatorial optimization problem. Let $S \subseteq V$ with $|S| = k$ be the set of nodes removed simultaneously, where k is the operational budget. After removal, surviving

organizations form new alliances according to the logistic model. The objective is:

$$S^* = \arg\min_{S \subseteq V,\, |S|=k} \mathrm{LCC}\Big(\mathrm{Rewire}(G \setminus S,\ \hat{P},\ \tau)\Big) \tag{9}$$

The parameter $\alpha \in \{0.25, 0.50, 0.75\}$ represents rewiring capacity, the fraction of lost edges the network can replace. Low α models a hostile enforcement environment and high α models a permissive one. Algorithm 3 describes the procedure.

Algorithm 3 Optimal targeting under rewiring

Require: G_{alliance}, G_{signed}, logistic model $\hat{P}$, budget k, rewiring capacity α

```
for each candidate set S ⊆ V with |S| = k do
    G' ← G_alliance \ S                          ▷ remove target nodes and incident edges
    Prune isolates from G'
    E_lost ← E_original − E_G'
    Score all non-adjacent pairs in G' using P̂ with features from G' and G_signed
    Add the top ⌊α · E_lost⌋ predicted edges to G'
    LCC(S) ← size of largest connected component of G'
end for
return S* = arg min_S LCC(S)
```

For $k = 1$, there are 91 candidate removals and each evaluation involves scoring $\sim N^2/2$ non-edges, so we solve the problem exactly by exhaustive enumeration. We compare the optimal set against four baselines: top-k by degree, top-k by betweenness, degree-based with rewiring, and the mean over 200 random k-subsets.

4 Results

We report findings for each stage of the pipeline: structural characterization (Stage 1), link prediction (Stage 2), resilience (Stage 3), and optimal targeting under rewiring (Stage 4).

4.1 Stage 1: Structural Characterization

Table 3 reports centrality rankings. CJNG dominates both measures (degree 0.31, betweenness 0.27), followed by Sinaloa (0.23, 0.07). After these two, degree drops to 0.067, confirming that the network is concentrated around two hubs. Talibanes, a local organization, ranks second in betweenness (0.10) despite modest degree (0.04), occupying a disproportionate share of shortest paths between clusters. Figure 6 confirms that local groups overlap with national cartels on betweenness.

Table 3: Top 5 organizations by degree and betweenness centrality.

Organization	Type	Degree Centr.	Betweenness Centr.
CJNG	National Cartel	0.3111	0.2684
Sinaloa	National Cartel	0.2333	0.0680
Chapitos	Faction	0.0667	0.0225
Unión Tepito	Local	0.0667	0.0572
Talibanes	Local	0.0444	0.1041

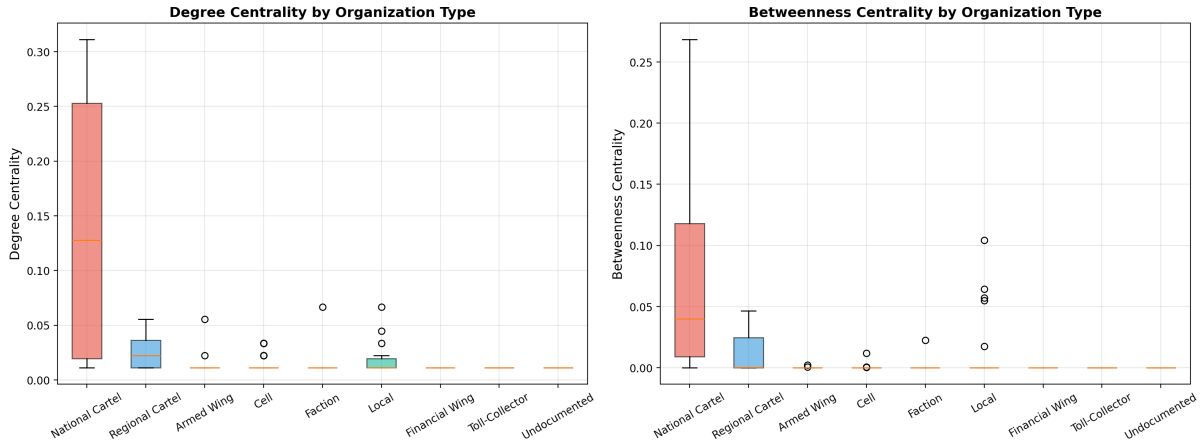

Figure 6: Distribution of degree and betweenness centrality by organization type.

Table 4 tests whether this structure is systematic. Both null models produce $C = 0$ (no triangles), while the real network has $C = 0.036$, indicating that triadic closure is present and not an artifact of random wiring. The real network is more compact ($L = 3.30$ vs. 6.75 ER, 4.48 BA) due to hub dominance, yet more fragmented (LCC = 50 vs. 62 ER, 91 BA), reflecting geographic and factional divisions.

Table 4: Comparison of empirical alliance network with ER and BA null models.

Network	Nodes	Edges	Clustering C	Avg. Path L	LCC Size
Real (Alliance)	91	91	0.0356	3.30	50
Erdős–Rényi	91	77	0.0000	6.75	62
Barabási–Albert	91	90	0.0000	4.48	91

The degree distribution (Figure 7) has a long right tail (CJNG at degree 28, Sinaloa at 21, then a sharp drop) that neither null model reproduces. With only 91 nodes a rigorous power-law fit is not possible (Clauset et al., 2009), but the distribution is heavy-tailed and hub-dominated, making the network vulnerable to targeted attack.

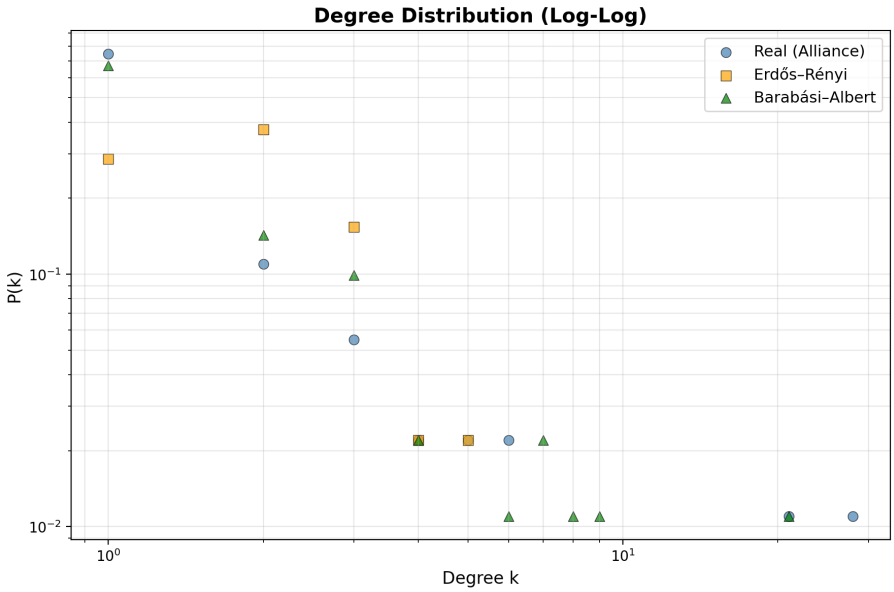

Figure 7: Degree distribution on log-log axes: real alliance network vs. ER and BA null models.

4.2 Stage 2: Link Prediction

All non-adjacent pairs are scored under the four models. Note that Jaccard and Adamic-Adar produce similarity scores rather than calibrated probabilities, and provide no intrinsic decision threshold. Given the sparsity of the alliance network, most pairs share no common neighbors, so both metrics assign a score of zero to the vast majority of non-edges.

Tables 5 and 6 report the logistic regression coefficients. The topology-only model (pseudo $R^2 = 0.076$) has only DegreeProduct as significant ($p < 0.001$). The full model (pseudo $R^2 = 0.144$) nearly doubles explanatory power: SameState dominates ($\beta = 1.73$, OR $= 5.66$, $p < 0.001$), DegreeProduct remains significant, and SharedRivals ($p = 0.062$) and BalanceScore ($p = 0.070$) are marginally significant. SharedAllies turns negative once geography is controlled for, and Complementarity is not significant ($p = 0.97$).

Table 5: Topology-only logistic model (Equation 7).

Feature	**Coef.**	**Std. Err.**	**Odds Ratio**	**p-value**
Intercept	-4.106	0.128	0.02	< 0.001***
SharedAllies (x_1)	0.301	0.231	1.35	0.193
DegreeProduct (x_6)	0.036	0.005	1.04	< 0.001***

Note: McFadden pseudo $R^2 = 0.076$. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 6: Full logistic model (Equation 8).

Feature	**Coef.**	**Std. Err.**	**Odds Ratio**	p**-value**
Intercept	-4.418	0.159	0.01	< 0.001***
SharedAllies (x_1)	-0.272	0.299	0.76	0.363
SharedRivals (x_2)	0.516	0.277	1.67	0.062
Complementarity (x_3)	0.013	0.315	1.01	0.967
SameState (x_4)	1.734	0.264	5.66	< 0.001***
BalanceScore (x_5)	0.359	0.198	1.43	0.070
DegreeProduct (x_6)	0.036	0.006	1.04	< 0.001***

Note: McFadden pseudo $R^2 = 0.144$. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 8 presents the score distributions across all four models. The topological metrics yield non-zero values for only a small fraction of dyads, reflecting the sparsity of shared neighborhoods. The logistic models, by contrast, produce continuous probability distributions over the full set of non-adjacent pairs.

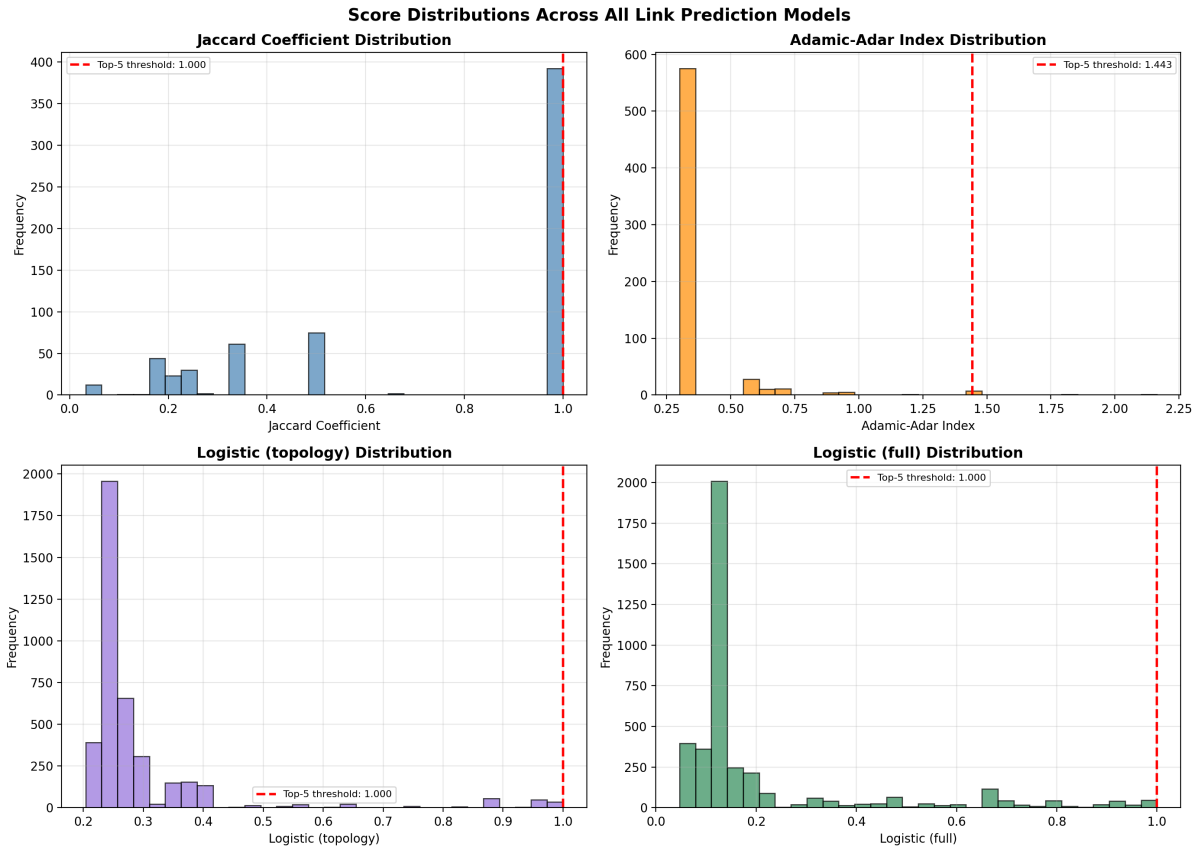

Figure 8: Score distributions across all four link prediction models. Red dashed lines mark the top-5 threshold for each model.

Table 7 compares the top-5 predicted pairs. Topological metrics surface peripheral satellites sharing a single ally (Jaccard = 1.0 means identical neighbor sets). Both logistic models instead predict alliances between major actors (Unión Tepito–Sinaloa, Chapitos–CJNG, Noreste–

CJNG), reflecting preferential attachment and geographic signals that topological metrics miss.

Table 7: Top 5 predicted alliances by each model.

Model	Organization A	Type	Organization B	Score
Jaccard				
	Nueva Plaza	Local	Guera	1.000
	Nueva Plaza	Local	Rusos	1.000
	Nueva Plaza	Local	Salazares	1.000
	Nueva Plaza	Local	Emperatriz	1.000
	Nueva Plaza	Local	Poniente o Laguna	1.000
Adamic-Adar				
	Golfo	Regional Cartel	Zetas VE	2.164
	Talibanes	Local	Pelones	1.821
	Viagras	Local	CJNG	1.443
	Reina del Sur	Undocumented	Juan L. (Moco)	1.443
	Unión Tepito	Local	Sindicato Libertad	1.443
Logistic (topology)				
	Unión Tepito	Local	Sinaloa	1.000
	Chapitos	Faction	CJNG	1.000
	Noreste	Regional Cartel	CJNG	1.000
	Noreste	Regional Cartel	Sinaloa	1.000
	Golfo	Regional Cartel	CJNG	1.000
Logistic (full)				
	Unión Tepito	Local	Sinaloa	1.000
	Chapitos	Faction	CJNG	1.000
	Noreste	Regional Cartel	CJNG	1.000
	Noreste	Regional Cartel	Sinaloa	1.000
	Golfo	Regional Cartel	CJNG	1.000

Table 8 reports the cross-validation results that adjudicate between these approaches. Jaccard (AUC = 0.47) and Adamic-Adar (0.48) perform at chance level, and the topology-only logistic model (0.41) performs below chance. Only the full model (0.73) achieves meaningful discriminative power, with the improvement attributable to SameState and SharedRivals. These results indicate that alliance formation in this network is governed primarily by geographic proximity and shared enmity rather than by existing alliance topology. Accordingly, the full model is the specification carried forward into Stage 4 for post-disruption rewiring prediction.

Table 8: 5-fold cross-validation AUC for link prediction models.

Model	Mean AUC	Std AUC
Jaccard	0.472	0.036
Adamic-Adar	0.484	0.031
Logistic (topology)	0.414	0.070
Logistic (full)	0.725	0.057

4.3 Stage 3: Network Resilience

We first remove CJNG (El Mencho), the node with the highest degree and betweenness centrality. Table 9 reports the structural impact. Removing CJNG drops the network from 91 to 67 nodes (23 former allies become isolates), with 31% of edges lost and the LCC halved from 50 to 26. Density and clustering increase because the loosely connected periphery was pruned, leaving a smaller, denser core (Figure 9).

Table 9: Alliance network metrics before and after removal of CJNG.

Metric	**Before**	**After**	Δ
Nodes	91	67	-24
Edges	91	63	-28
Density	0.0222	0.0285	$+0.0063$
Components	8	10	$+2$
LCC Size	50	26	-24
Diameter	8	3	-5
Avg. Path Length	3.30	2.14	-1.15
Clustering C	0.0356	0.0632	$+0.0276$

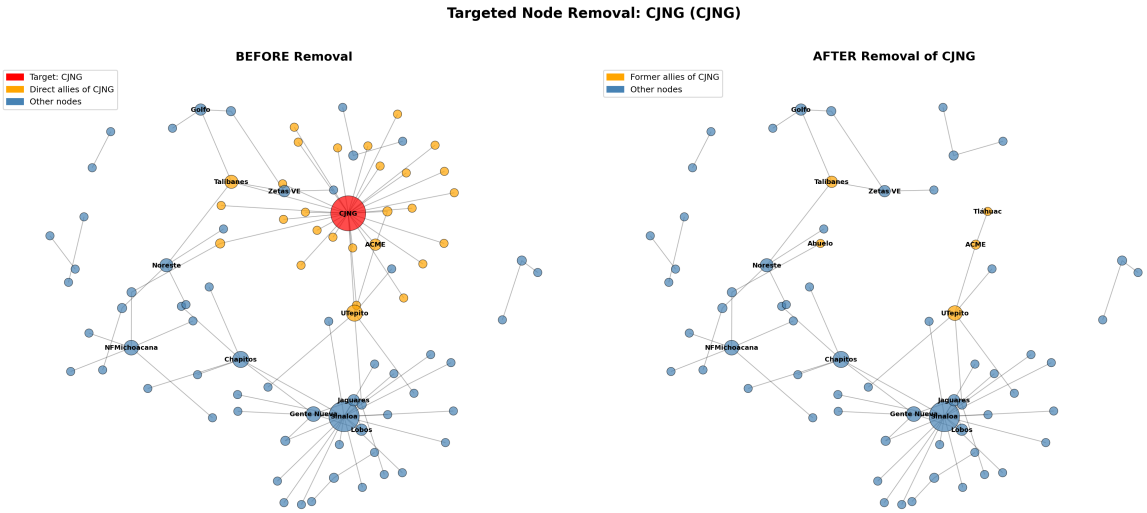

Figure 9: Alliance network before and after targeted removal of CJNG (El Mencho). Orange nodes are former CJNG allies; the red node in the left panel is CJNG itself.

Extending this to sequential removal, Figure 10 traces the fragmentation curve under iterative removal of the top-5 hubs by degree. The first removal (CJNG) causes the largest drop; the curve flattens after two or three removals, consistent with the theoretical prediction for heavy-tailed networks (Albert et al., 2000). The real-world enforcement sequence of 2024–2026, targeting El Mencho, El Mayo, and Los Cuinis leadership, follows the order our adaptive algorithm prescribes.

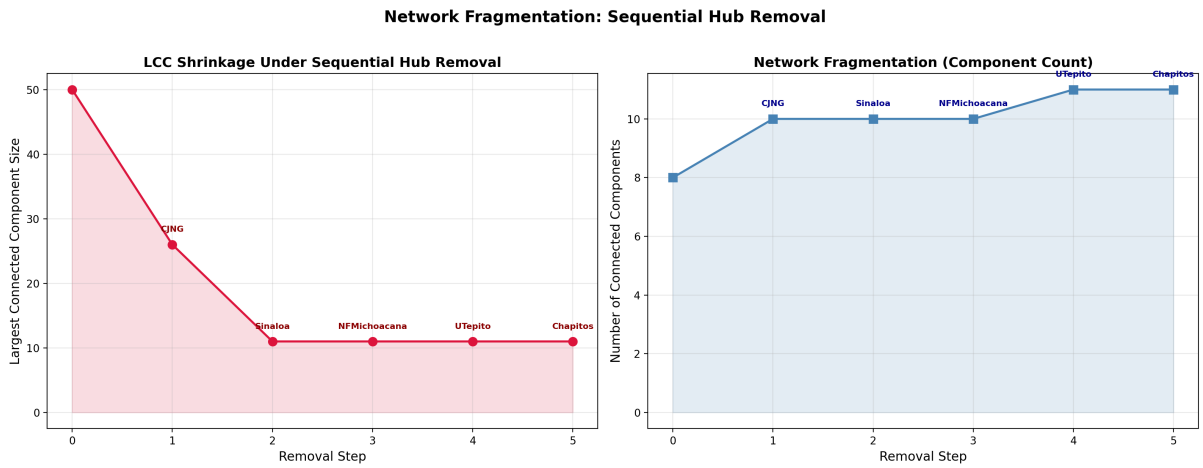

Figure 10: Network fragmentation under sequential targeted attack. Left: LCC size after each removal. Right: number of connected components.

4.4 Stage 4: Targeting and Rewiring

After removing CJNG, we recompute all 6 dyadic features on the 67-node surviving network and score every non-adjacent pair with the full logistic model. Figure 11 shows the top-10 predicted new alliances.

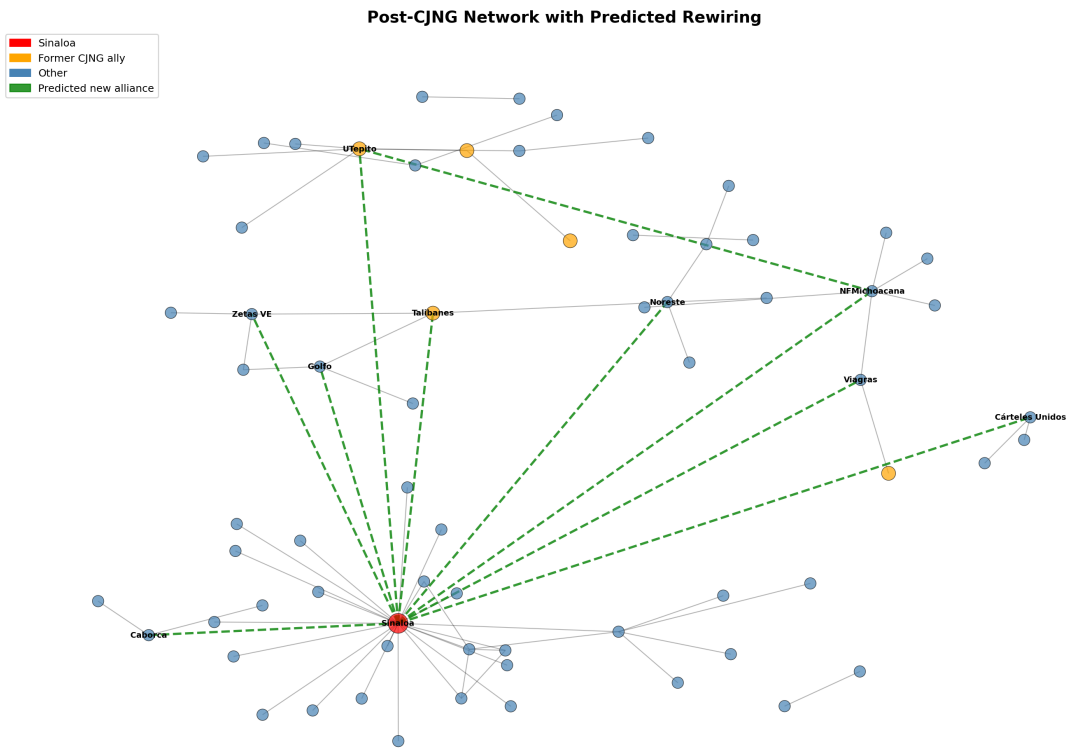

Figure 11: Surviving network after CJNG removal. Existing edges in gray; top-10 predicted new alliances as dashed green edges. Orange = former CJNG allies; red = Sinaloa.

The key question is whether former CJNG satellites are absorbed by Sinaloa (consolidation) or form independent clusters (fragmentation). The simulation identifies what is structurally most plausible, not what will occur. The optimal targeting analysis (Equation 9) solves for $k = 1$ across three rewiring capacities ($\alpha = 0.25, 0.50, 0.75$). Figure 12 compares the optimal target against degree, betweenness, and random baselines.

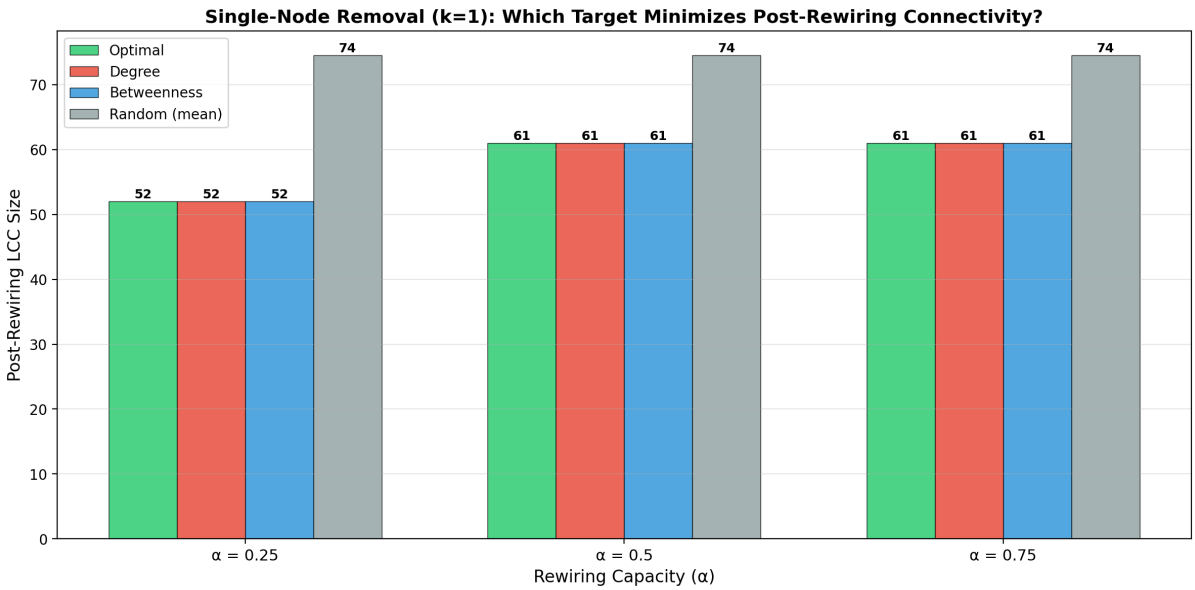

Figure 12: Post-rewiring LCC size by targeting strategy and budget k, across rewiring capacity levels α. Lower is better (smaller surviving cooperation structure).

If the optimal target differs from the highest-degree node, static centrality rankings are misleading once reorganization is accounted for. At low α (harsh enforcement) the strategies may converge; at high α (permissive environment) the gap widens as the network compensates for disruption.

4.5 Summary of Findings

The four stages yield a consistent picture. The alliance network is sparse, hub-dominated, and clustered above random baselines (Stage 1). Topology alone cannot predict alliance formation, as geography and shared enemies are the primary drivers, with the full logistic model achieving AUC 0.73 against chance-level performance for all other models (Stage 2). Removing CJNG eliminates 31% of edges and halves the largest component (Stage 3), and the sequential fragmentation curve matches theoretical predictions for heavy-tailed networks. When the surviving network is allowed to reorganize, the logistic model predicts rewiring patterns that may partially restore connectivity, raising the question of whether static targeting strategies remain optimal (Stage 4).

5 Discussion

The results paint a bipolar cooperation structure held together by two national hubs. CJNG and Sinaloa account for over half the alliance ties among non-isolate nodes, and their rivalry (weight 7, the dataset maximum) defines the principal axis of conflict. Most smaller organizations are satellites of one hub or the other. Yet degree is not the only dimension that matters: local organizations such as Talibanes and Unión Tepito rank high on betweenness despite few alliances, occupying bottleneck positions whose disruption would fragment communication channels in ways that degree alone does not predict (Duijn et al., 2014). Cross-state alliances link production regions to distribution corridors, partly mapping supply-chain relationships, while same-state alliances reflect territorial cooperation. The logistic model confirms this pattern, with SameState as the dominant predictor of alliance formation (OR = 5.66, $p < 0.001$).

The link prediction results carry a methodological implication. In sparse, hub-dominated networks, topological similarity metrics perform at chance because most pairs share zero neighbors. Contextual features such as geography and shared enemies carry the predictive signal. This is consistent with Berlusconi et al. (2016), who found that attribute-enriched models outperformed topology-only baselines in Italian Mafia networks. The practical consequence is that intelligence analysts predicting future alliances should prioritize geographic proximity and rivalry structure over alliance topology.

The node-removal simulations quantify what the centrality analysis implies. Removing CJNG halves the largest component and isolates 23 satellite organizations; sequential removal of the top three hubs captures most of the achievable fragmentation. This matches the theoretical prediction for heavy-tailed networks (Albert et al., 2000) and aligns with the real-world enforcement strategy observed between 2024 and 2026. However, Calderón et al. (2015) documented that decapitation creates power vacuums that produce violence. Our simulations quantify structural damage but do not capture the violence that fills the vacuum. Fragmentation is not pacification.

Events since 2020 track the structural vulnerabilities identified in our analysis. Ismael "El Mayo" Zambada, co-founder of Sinaloa (the second-highest degree node), was arrested in Texas on July 25, 2024, triggering a civil war between Los Chapitos and Zambada loyalists. In network terms, one high-degree node split into competing lower-degree nodes with alliance edges replaced by rivalry edges. On February 22, 2026, Mexican military forces killed Nemesio Oseguera Cervantes ("El Mencho"), CJNG's leader and the highest-degree, highest-betweenness node in the 2020 network. Our simulation predicts that removing CJNG drops the LCC from 50 to 26 and turns 23 satellites into isolates. The immediate retaliatory attacks in Jalisco are consistent with hub removal in a centralized network. In February 2025, the US designated both CJNG and Sinaloa as Foreign Terrorist Organizations, and mass extraditions (29 figures in February, 26 more in August 2025) dismantled Los Cuinis, CJNG's strongest ally, before El Mencho's elimination. The factions created by Zambada's arrest did not reconcile; the network did not heal.

6 Conclusion

This paper analyzed the BACRIM 2020 signed network of Mexican criminal organizations through a four-stage pipeline of structural characterization, link prediction, targeted-removal simulation, and optimal targeting under rewiring. The alliance network is sparse, clustered above random baselines, and dominated by two hubs whose removal fragments the cooperation structure. Topology alone cannot predict alliance formation. Geography and shared enemies are the primary drivers, and only a model incorporating these contextual features achieves meaningful predictive performance. Targeted hub removal produces fragmentation consistent with theoretical predictions for heavy-tailed networks, and post-disruption rewiring simulations identify which new alliances are structurally most plausible. Events between 2024 and 2026 confirm that the nodes flagged as most central were the ones law enforcement targeted, and that their removal produced consequences the simulation anticipated.

Several limitations qualify these findings. The 2020 snapshot cannot capture how ties form or dissolve in response to enforcement actions, and the rewiring simulation assumes the same formation mechanisms apply after disruption. The data is necessarily incomplete, as covert relationships may go unrecorded and organizational boundaries are sometimes blurry (Sparrow, 1991). We treat each organization as a unitary actor, but internal factions (e.g., Los Chapitos vs. Los Mayos) may have distinct alliance structures, as the 2024 Sinaloa civil war demonstrated. All alliance edges are treated as equal, though the original weight data (number of active states) could proxy for tie strength and might produce different centrality rankings.

These limitations point toward productive extensions. Temporal analysis over multiple BACRIM snapshots would allow direct testing of link predictions. ERGMs or STERGMs could formalize the same hypotheses with richer generative assumptions, and signed-link prediction (forecasting both alliances and rivalries) is a natural next step. The optimal targeting framework could be extended to sequential multi-round operations with inter-round rewiring, better reflecting the iterative nature of real enforcement campaigns. Despite these open questions, a single static snapshot analyzed with standard network tools and a dyadic formation model identified where the network was most fragile and how it would reorganize, information that proved consistent with the disruption events that followed.

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